

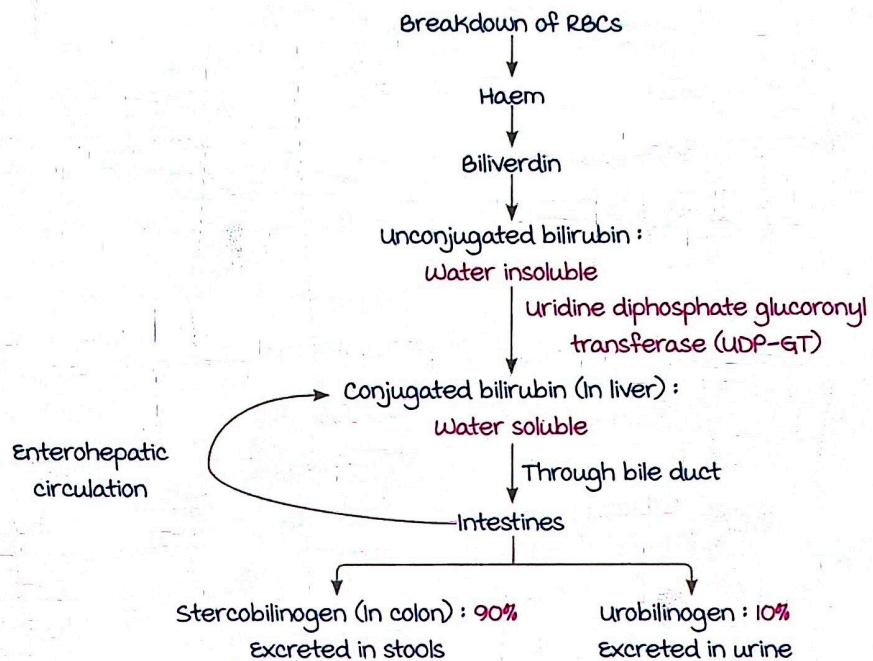
NEONATAL JAUNDICE

Incidence :

Seen in upto 60% newborns

- Pathological : 5-10% (Require treatment).
- Physiological : majority (No treatment required).

Bilirubin metabolism :



Assessment of jaundice :

1. Serum bilirubin levels : **Best investigation.**

- Bilirubin values : Total serum bilirubin (TSB)
 - Direct : Conjugated
 - Indirect : Unconjugated

2. Screening of baby :

Initial clinical assessment.

- **Kramer method** : Visual assessment.

	Kramer zones	Appropriate serum bilirubin
	Face & neck (Seen first in the eyes)	5-7 mg/dL
	Chest & upper abdomen	7-9 mg/dL
	Lower abdomen & thighs	9-11 mg/dL
	Legs, arms & forearms	11-13 mg/dL
	Palms & soles (Danger sign)	>13-15 mg/dL → ↑ risk of kernicterus (brain injury)

Neonatal Jaundice

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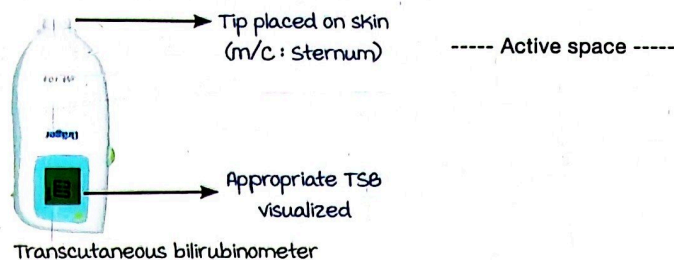
- Trans cutaneous bilirubinometry (T_{CB}):

Better than visual assessment.

Unreliable in:

- High bilirubin levels (>15 mg/dL).
- Preterm babies.

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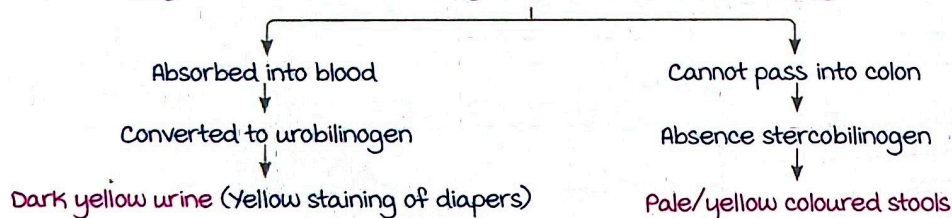
Types Of Jaundice

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	Physiological jaundice	Pathological jaundice
Incidence	Common	Less common
Bilirubin:		
a. Type	Always conjugated	Can be conjugated or unconjugated
b. values (TSB)	mildly elevated	Largely elevated: >15 mg/dL
c. Rate of increase (TSB)	-	>5 mg/dL
Time of onset	Usually 24 to 72 hours of age	<24 hours of age
Duration	1-2 weeks	Beyond 3-4 weeks
Peak levels by	3 rd to 5 th day	-
Palms & soles	Not affected	Affected
Treatment	Not required: Wait & watch	Required

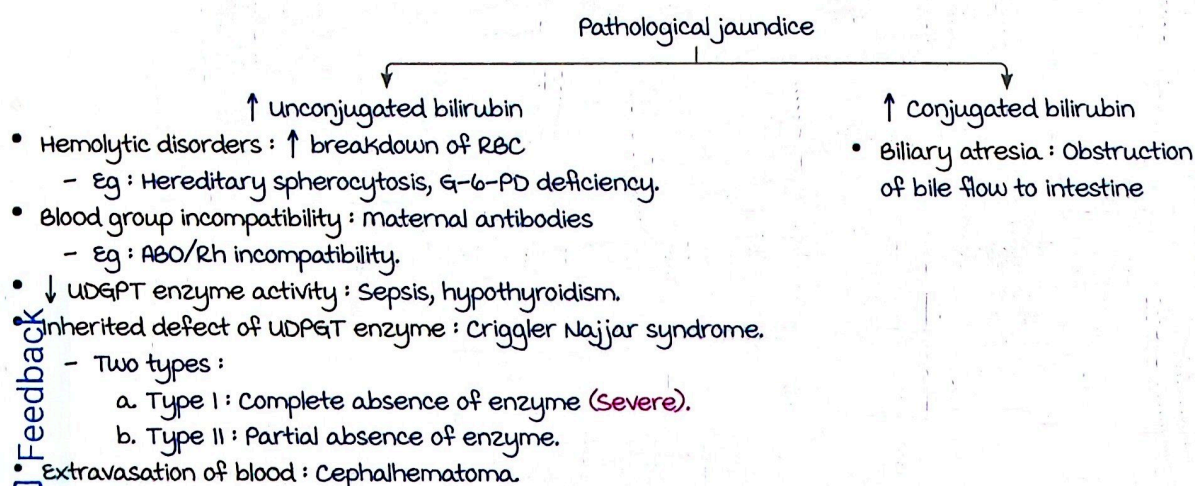
Note: Clay/pale coloured stools indicate obstruction.

- Can be in the bile duct or intestine.
- $\uparrow$ conjugated bilirubin (Pathological jaundice): value >1 mg/dL or $>20\%$ of TSB.



Causes Of Jaundice

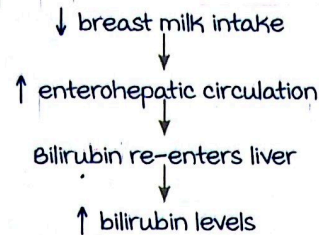
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Breast feeding jaundice :AKA **suboptimal intake hyperbilirubinemia.**

Pathogenesis :



Onset : 5-7 days of age (upto 1 week).

Clinical features : Excessive weight loss (**>10% of body weight**).

management : Increase breast feeding frequency and duration.

Breastmilk jaundice

- Jaundice **>3 weeks**.
- Diagnosis of **exclusion**.

Pathogenesis :

 β -glucuronidase in milk $\rightarrow$ $\downarrow$ conjugation of bilirubin $\rightarrow$ $\uparrow$ unconjugated bilirubin levels.

management :

- **Continue breastfeeding**.
- Transient & self resolving.

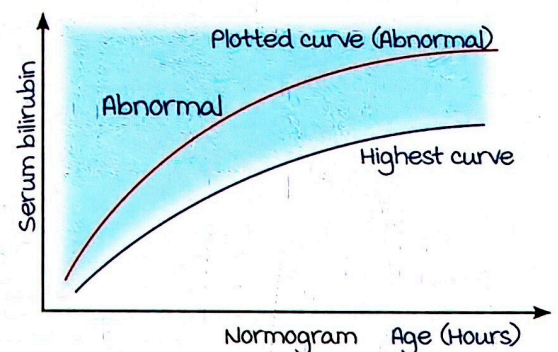
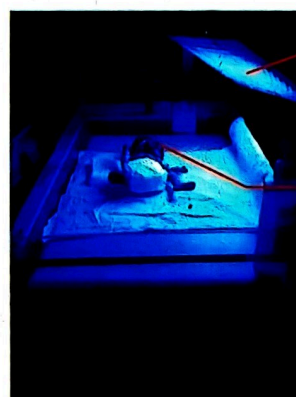
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Treatment of Pathological Jaundice

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NORMOGRAM :

used to decide course of treatment of jaundice.

**TREATMENT METHODS :****Phototherapy :**

Blue green LED (460-490 nm)

Protection of eye & gonads (Testis)

Phototherapy

• mechanism :

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- Structural isomerism** : Bilirubin structure changed.
Bilirubin $\rightarrow$ Lumirubin $\rightarrow$ Excreted.
 - Photoisomerisation** : Light $\rightarrow$ Bilirubin isomers formed.
 - Photooxidation**.
- minimum irradiance : 30 microwatt/cm²/nm.
 - minimum distance from light source : 30-45 cm.
 - Termination of phototherapy : Atleast 2 consequent values within the normal curves on normogram.
 - C/I : Conjugated bilirubin (Brown pigment formation $\rightarrow$ **bronze baby syndrome**).

Double volume exchange transfusion (DVET) :

Indication :

- very high bilirubin level (Abnormal in normogram).
- Rh isoimmunization with
 - $\rightarrow$ Cord blood bilirubin >5 mg/dL or
 - $\rightarrow$ Cord blood hemoglobin <10 mg/dL.

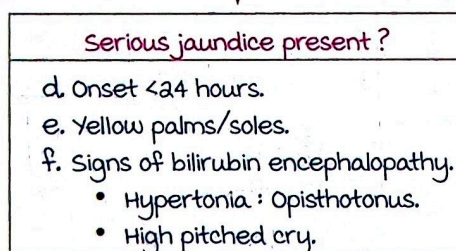
mechanism : Simultaneous removal of blood with $\uparrow$ bilirubin & replacement with equal volume of fresh blood.

Amount of blood :

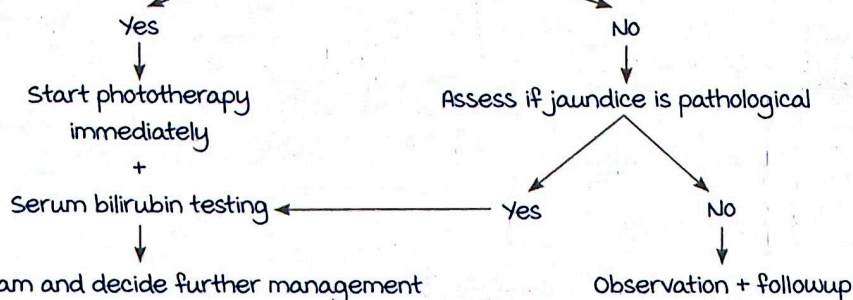
- Normal amount of blood in the body : 80-90 mL/kg.
- volume exchanged : 160-180 mL/kg.

MANAGEMENT PROTOCOL

Newborn jaundice screening
Every 12 hours for 3-5 days post-birth



Opisthotonus



Feedback